

Datasets (basic)

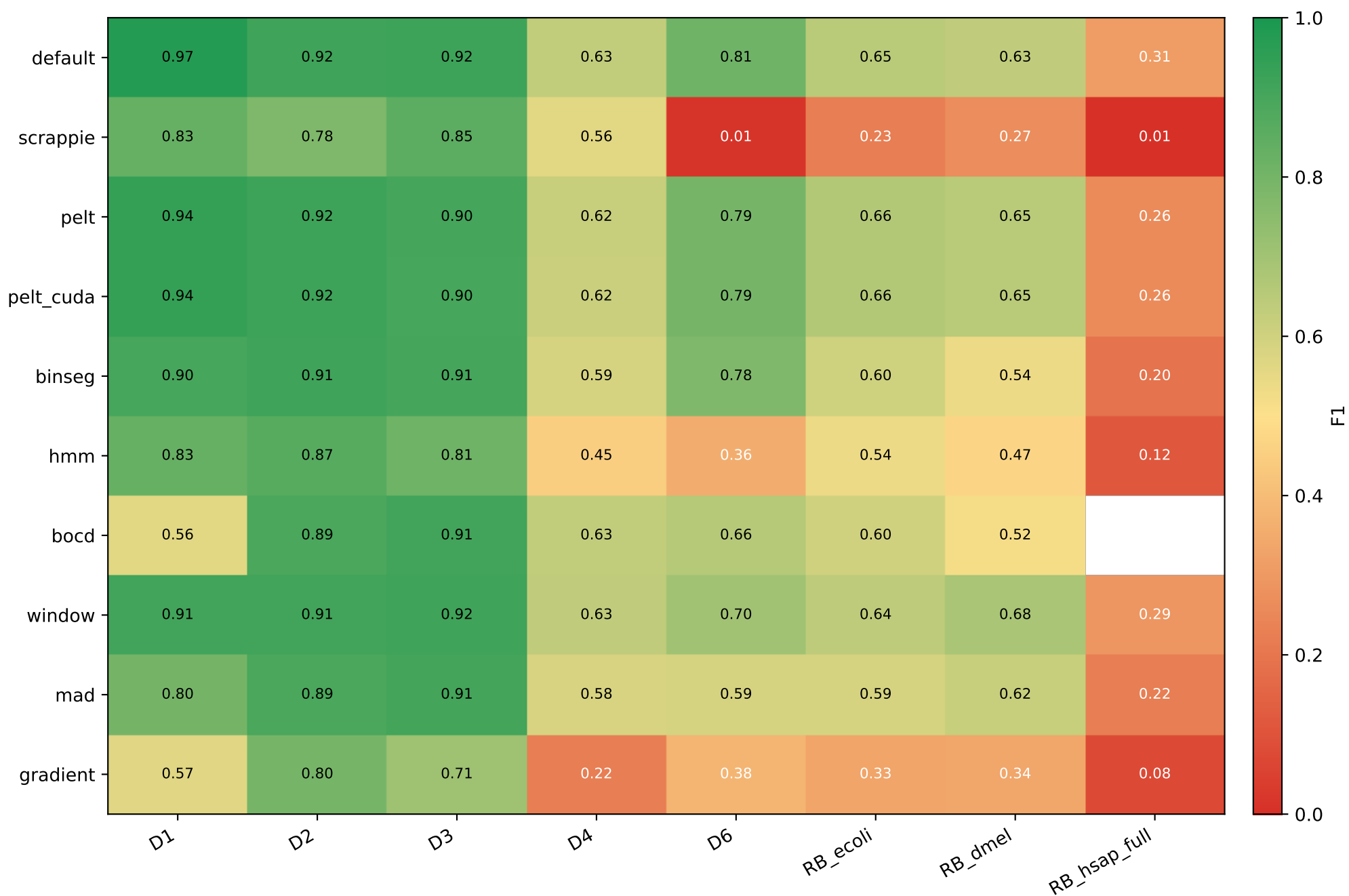
Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh38)	R10.4.1	12,000	3.0 GB	7.10 GB

Datasets — extended characterization

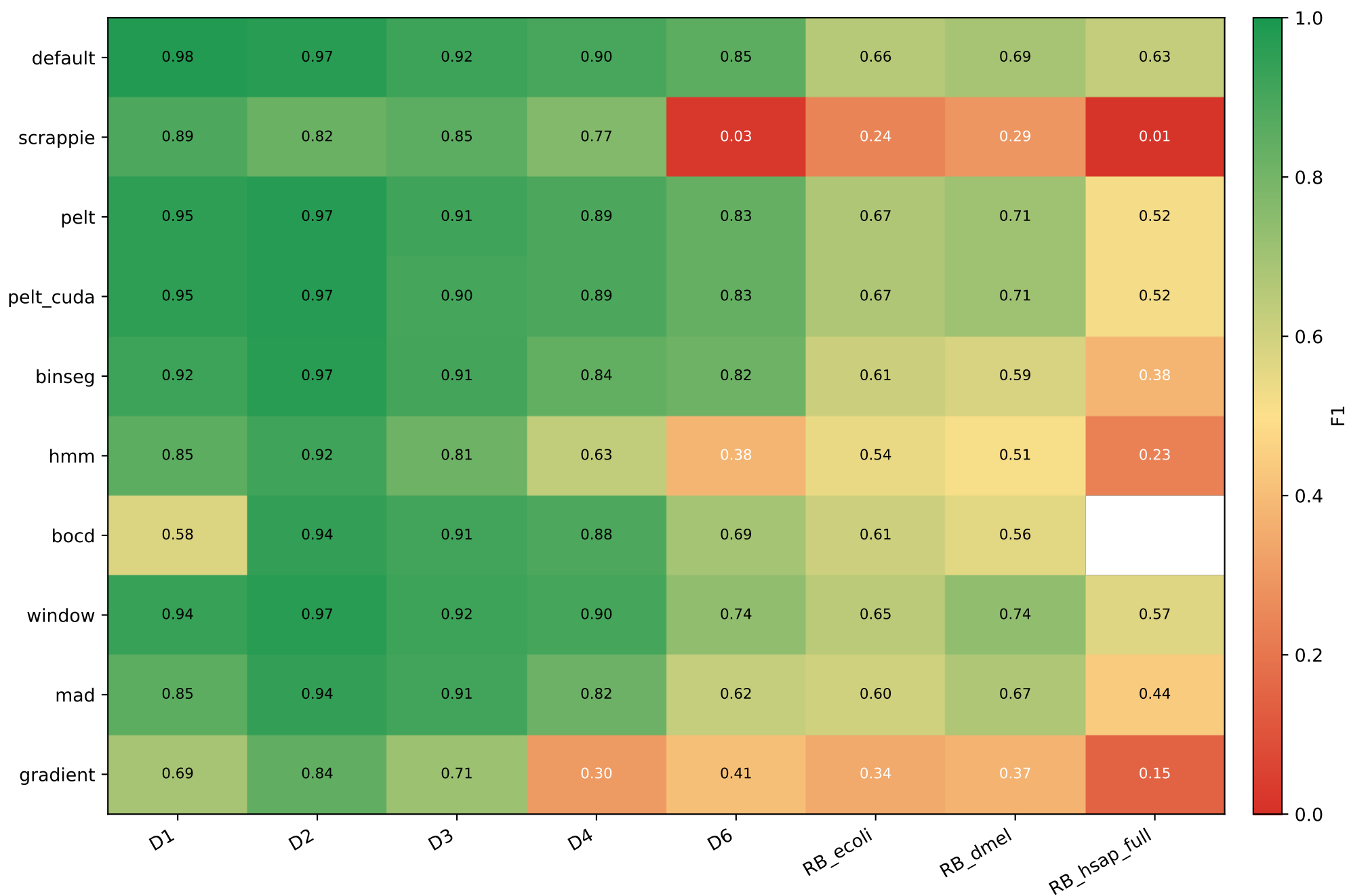
DS	Organism	Chem	Reads	Ref bp	GC%	contigs	N50 (Mb)	N%	masked%
D1	SARS-CoV-2	R9.4	1,383	0.03 Mb	38.0	1	0.03	0.00	0.0
D2	E. coli	R9.4	89	5.23 Mb	50.5	1	5.23	0.00	0.0
D3	S. cerevisiae	R9.4	500	12.16 Mb	38.1	17	0.92	0.00	0.0
D4	C. reinhardtii	R9.4	500	111.10 Mb	64.1	53	7.78	3.65	32.4
D6	E. coli	R10.4	294	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_ecoli	E. coli	R10.4.1	12,000	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_dmel	D. melanogaster	R10.4.1	12,000	143.73 Mb	42.0	1,870	25.29	0.80	20.9
RB_hsap_fullH	sapiens (GRCh38)	R10.4.1	12,000	3.21 Gb	41.0	455	145.14	4.98	49.5

Ref bp = total reference length; *N50* = contig N50; *N%* = ambiguous N bases; *masked%* = soft-masked low-complexity / repeat content (lowercase in FASTA).

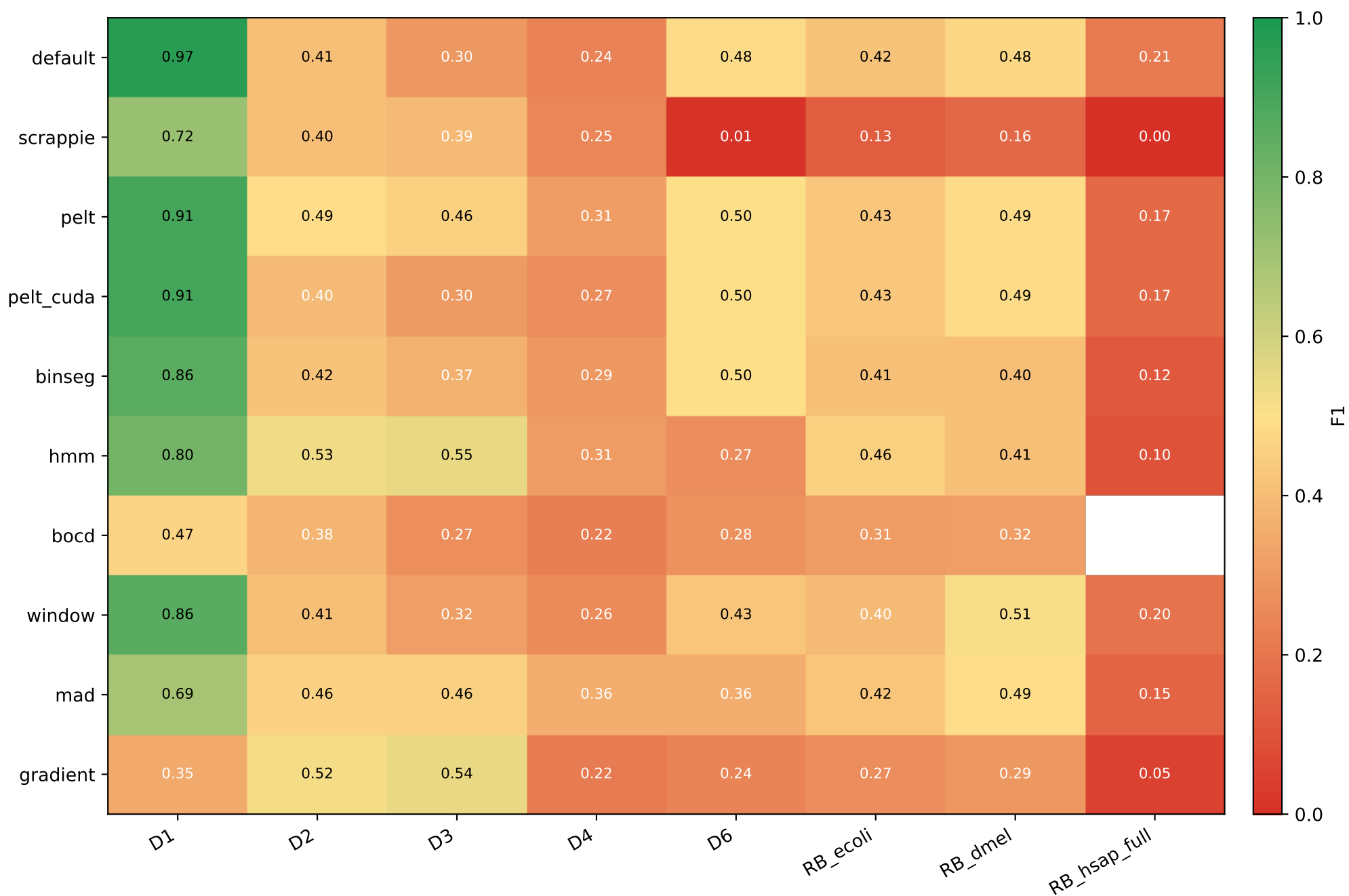
Primary: F1 with $|\text{center_q} - \text{center_t}| \leq 10 \text{ kb}$



Lenient: F1 with $|\text{center_q} - \text{center_t}| \leq 100 \text{ kb}$



Setup A reference: F1 with truth-overlap ≥ 0.1



Per-dataset F1 across 4 metrics — best per column highlighted

D1 (truth=1,381,999 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.975	0.979	0.968	0.920
scrappie	0.831	0.886	0.720	0.477
pelt	0.938	0.953	0.906	0.671
pelt_cuda	0.938	0.953	0.906	0.671
binseg	0.901	0.921	0.856	0.595
hmm	0.831	0.849	0.802	0.753
bocd	0.560	0.578	0.471	0.278
window	0.914	0.937	0.860	0.610
mad	0.799	0.851	0.689	0.431
gradient	0.566	0.689	0.346	0.175

D2 (truth=350,620 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.915	0.968	0.410	0.071
scrappie	0.777	0.821	0.404	0.066
pelt	0.919	0.972	0.492	0.084
pelt_cuda	0.919	0.972	0.397	0.066
binseg	0.915	0.968	0.417	0.070
hmm	0.866	0.917	0.530	0.096
bocd	0.894	0.945	0.378	0.034
window	0.914	0.967	0.410	0.069
mad	0.893	0.945	0.462	0.075
gradient	0.797	0.844	0.518	0.090

D3 (truth=445 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.918	0.920	0.295	0.019
scrappie	0.853	0.855	0.392	0.020
pelt	0.903	0.915	0.456	0.036
pelt_cuda	0.902	0.904	0.304	0.024
binseg	0.906	0.908	0.369	0.021
hmm	0.810	0.813	0.545	0.046
bocd	0.911	0.914	0.271	0.019
window	0.916	0.918	0.319	0.040
mad	0.909	0.911	0.460	0.033
gradient	0.708	0.711	0.543	0.040

D4 (truth=485 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.632	0.897	0.237	0.039
scrappie	0.559	0.769	0.247	0.021
pelt	0.619	0.891	0.309	0.041
pelt_cuda	0.617	0.888	0.267	0.035
binseg	0.589	0.839	0.295	0.022
hmm	0.452	0.630	0.307	0.016
bocd	0.629	0.884	0.224	0.020
window	0.633	0.902	0.260	0.039
mad	0.584	0.819	0.362	0.023
gradient	0.224	0.303	0.217	0.026

Per-dataset F1 across 4 metrics — best per column highlighted

D6 (truth=1,167,335 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.811	0.851	0.485	0.077
scrappie	0.014	0.026	0.009	0.003
pelt	0.793	0.833	0.498	0.083
pelt_cuda	0.793	0.833	0.498	0.083
binseg	0.776	0.815	0.496	0.085
hmm	0.357	0.382	0.267	0.055
bocd	0.660	0.692	0.278	0.015
window	0.701	0.738	0.425	0.069
mad	0.589	0.622	0.362	0.056
gradient	0.382	0.407	0.239	0.036

RB_ecoli (truth=12,680 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.649	0.657	0.419	0.030
scrappie	0.225	0.239	0.131	0.016
pelt	0.663	0.673	0.429	0.031
pelt_cuda	0.663	0.673	0.429	0.029
binseg	0.604	0.612	0.413	0.030
hmm	0.536	0.544	0.456	0.064
bocd	0.600	0.606	0.307	0.012
window	0.637	0.647	0.398	0.030
mad	0.590	0.598	0.422	0.038
gradient	0.333	0.344	0.269	0.037

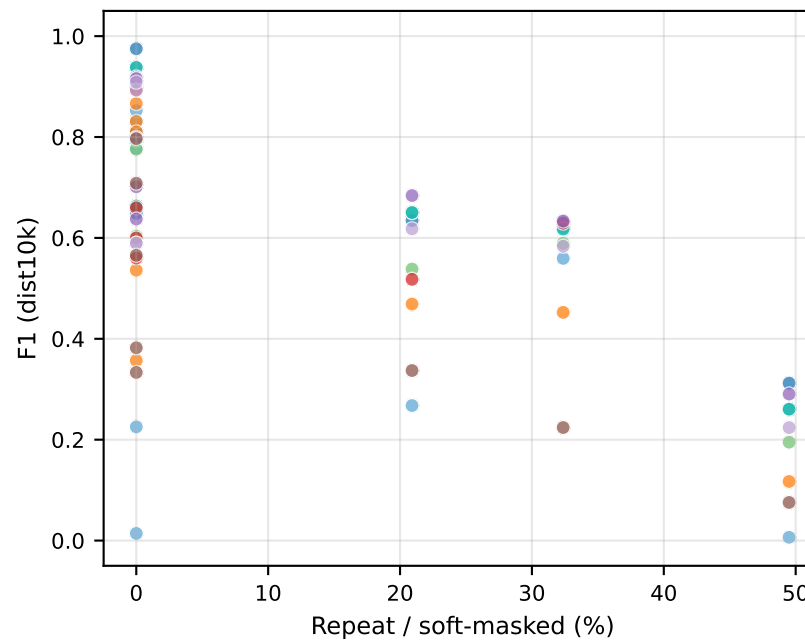
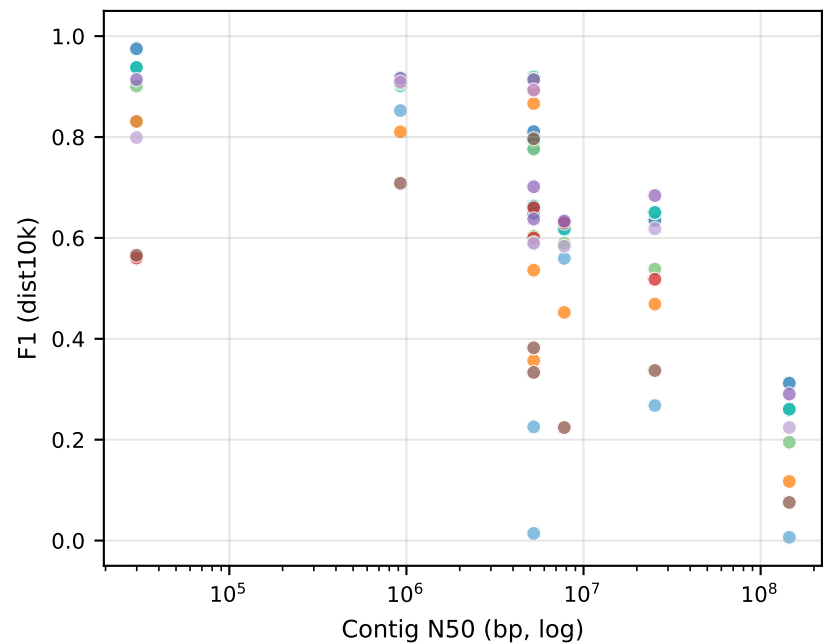
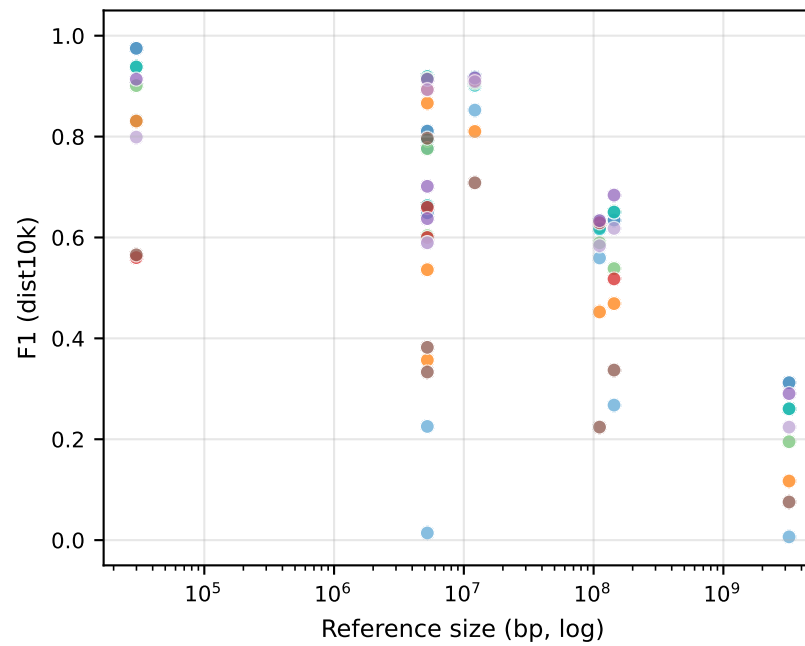
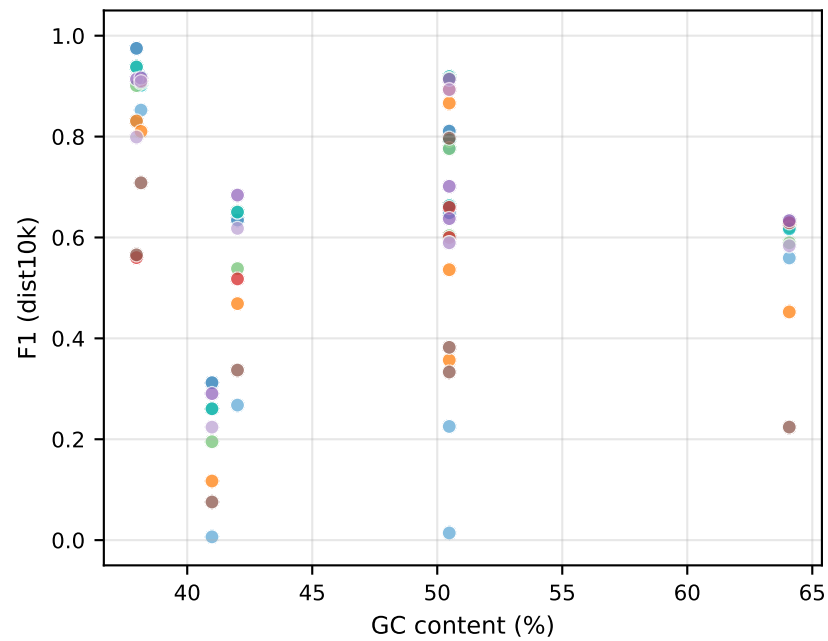
RB_dmel (truth=11,071 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.635	0.689	0.480	0.123
scrappie	0.268	0.291	0.162	0.027
pelt	0.651	0.708	0.491	0.127
pelt_cuda	0.650	0.707	0.491	0.127
binseg	0.538	0.586	0.404	0.088
hmm	0.469	0.514	0.411	0.115
bocd	0.518	0.562	0.315	0.034
window	0.684	0.742	0.510	0.141
mad	0.618	0.673	0.494	0.138
gradient	0.337	0.371	0.293	0.070

RB_hsap_full (truth=12,716 reads)

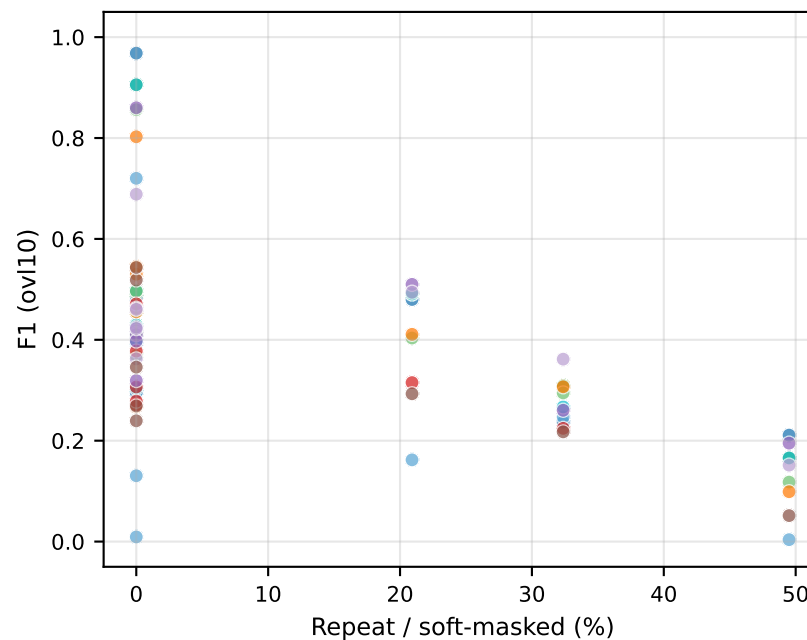
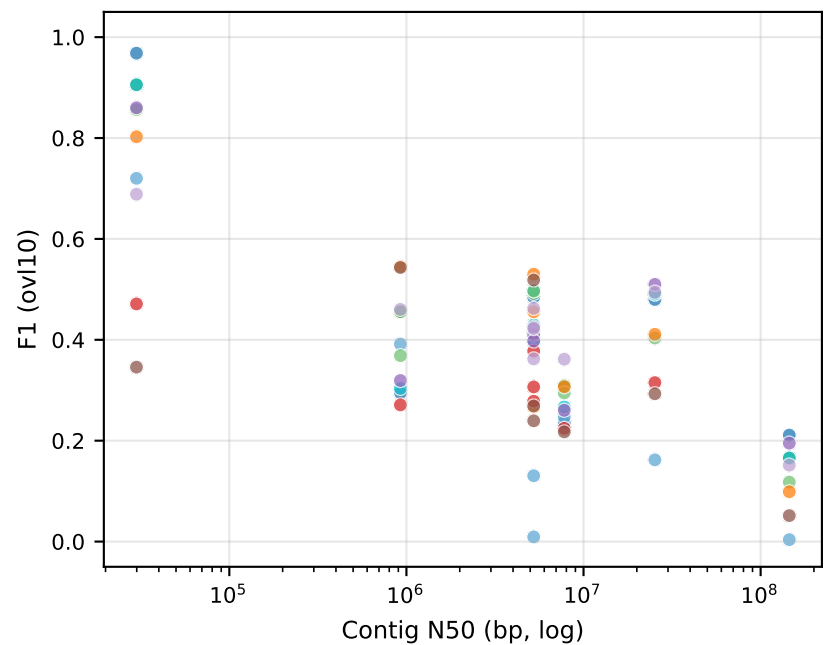
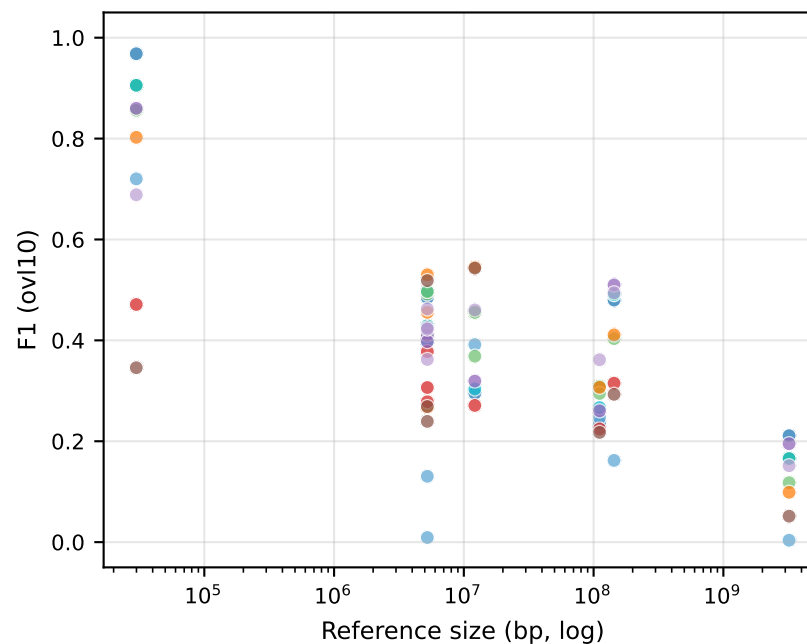
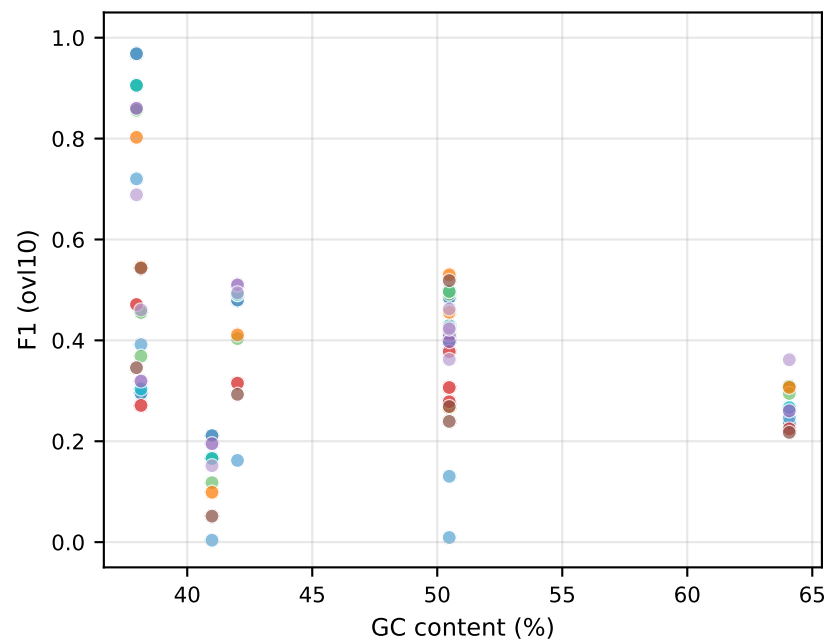
seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.312	0.625	0.211	0.062
scrappie	0.006	0.012	0.004	0.001
pelt	0.260	0.522	0.166	0.045
pelt_cuda	0.260	0.522	0.166	0.045
binseg	0.195	0.382	0.118	0.033
hmm	0.117	0.232	0.099	0.022
window	0.290	0.569	0.195	0.056
mad	0.224	0.442	0.151	0.042
gradient	0.076	0.146	0.051	0.013

F1 (dist10k) vs reference attributes — per segmenter



- default
- scrappie
- pelt
- pelt_cuda
- binseg
- hmm
- bocd
- window
- mad
- gradient

F1 (ovl10) vs reference attributes — per segmenter



- default
- scrappie
- pelt
- pelt_cuda
- binseg
- hmm
- bocd
- window
- mad
- gradient

Pearson correlation: F1 (dist10k) vs reference attribute



Pearson correlation: F1 (ovl10) vs reference attribute



CV-best HMM/BOCD config per dataset (3-fold read-level CV, ovl $\geq$ 0.1 metric)

Dataset	Segmenter	Best config	F1_test	$\pm$ std	#cfgs
D1	hmm	RH2_HMM_STATES=6 R	0.8675	0.0004	12
D1	bocd	RH2_BOCD_EXP_LEN=6	0.5068	0.0010	6
D2	bocd	RH2_BOCD_EXP_LEN=6	0.3877	0.0004	6
D2	hmm	RH2_HMM_STATES=4 R	0.5381	0.0015	12
D3	hmm	RH2_HMM_STATES=6 R	0.5583	0.0197	12
D4	bocd	RH2_BOCD_EXP_LEN=6	0.0034	0.0002	6
D4	hmm	RH2_HMM_STATES=5 R	0.0058	0.0008	12
D6	hmm	RH2_HMM_STATES=3 R	0.0162	0.0001	1
D6	bocd	RH2_BOCD_EXP_LEN=6	0.3187	0.0009	6
RB_dmel	hmm	RH2_HMM_STATES=6 R	0.4286	0.0064	12
RB_dmel	bocd	RH2_BOCD_EXP_LEN=6	0.3560	0.0105	6
RB_ecoli	bocd	RH2_BOCD_EXP_LEN=6	0.3566	0.0061	6
RB_ecoli	hmm	RH2_HMM_STATES=6 R	0.5040	0.0036	12

Speedup table — default vs tuned-pelt vs pelt_cuda

DS	default wall	pelt wall	pelt_cuda wall	pelt speedup	pelt_cuda speedup	F1 default	F1 pelt	F1 pelt_cuda
D1	1359.1s	10420.1s	1333.5s	0.13×	1.02×	0.975	0.938	0.938
D2	549.1s	—	722.8s	—	0.76×	0.915	0.919	0.919
D3	5.8s	—	4.0s	—	1.47×	0.918	0.903	0.902
D4	13.0s	89.7s	41.9s	0.15×	0.31×	0.632	0.619	0.617
D6	1331.8s	37353.9s	9226.0s	0.04×	0.14×	0.811	0.793	0.793
RB_ecoli	54.6s	613.9s	238.6s	0.09×	0.23×	0.649	0.663	0.663
RB_dmel	366.1s	1320.3s	1306.2s	0.28×	0.28×	0.635	0.651	0.650
RB_hsap_full	—	40754.1s	39664.2s	—	—	0.312	0.260	0.260